

Court Vision - NBA Analytics Platform

Problem

Domain: Sports Analytics - NBA Basketball Data Exploration and Analysis

Basketball analysts, coaches, and fans need to efficiently search and explore large amounts of NBA data including game results, player statistics, team performance metrics, and draft history. Raw datasets are difficult to navigate through SQL files or spreadsheets. Users lack an intuitive interface to:

- Search across game records and player statistics
- Compare team performance metrics
- Analyze draft history across years
- View detailed game information with player-level statistics
- Predict game outcomes based on team performance

Solution

Court Vision is a comprehensive sports analytics platform that provides an intuitive web interface for exploring NBA basketball data. The application transforms raw database records into actionable insights through organized, user-friendly pages backed by a MySQL database.

Key Features

- **Game Browsing & Details:** users can view recent games with comprehensive statistics for both teams and individual players

- **Team Rankings:** dynamic rankings based on wins and point differential with customizable filters
- **Draft Explorer**: full-text search across NBA draft history by year, player, and team
- **Matchup Predictor:** machine learning-based predictions for game outcomes using logistic regression trained on historical data
- **User Authentication:** secure login with BCrypt password hashing and profile mgmt

User Interface

1. Games List Page (`/games`)
    a. Displays recent NBA games (up to ~1,300 games)
    b. Shows matchups, dates, and team information
    c. Database query: retrieves games ordered by date with team stats
2. Game Detail Page (`/game/{gameId}`)
    a. Shows comprehensive game statistics for both teams
    b. Displays player-level box scores (points, rebounds, assists, shooting percentages)
    c. Database query: joins games with player game statistics, includes player biographical data
3. Team Rankings Page (`/rankings`)
    a. Ranks NBA teams by wins and point differential
    b. Filters by minimum games played (1 - 82 games)
    c. Database query: aggregates game results by team using GROUP BY and SUM
4. Draft Explorer Page (`/draft`)
    a. Search draft picks by year, player name, or team
    b. Shows draft information with player position and college data
    c. Database query: searches draft history with JOINs to player/draft info
5. Matchup Predictor Page (`/predictor`)
    a. Predicts game outcomes between selected teams
    b. Displays win probability percentage and projected point margin
    c. Database query: Loads team statistics for logistic regression model training
6. User Profile Page (`/profile`)
    a. Displays and allows editing of user account information
    b. Database query: Retrieves and updates user records from users table
7. Home Page (home)
    a. Authenticated user dashboard with personalized greeting
    b. Database query: Retrieves logged-in user information from users table
8. Non-database-driven UI pages:
    a. Login page
    b. Signup page
    c. Landing page

SpringBoot Maven Project 2 Style JDBC Connection

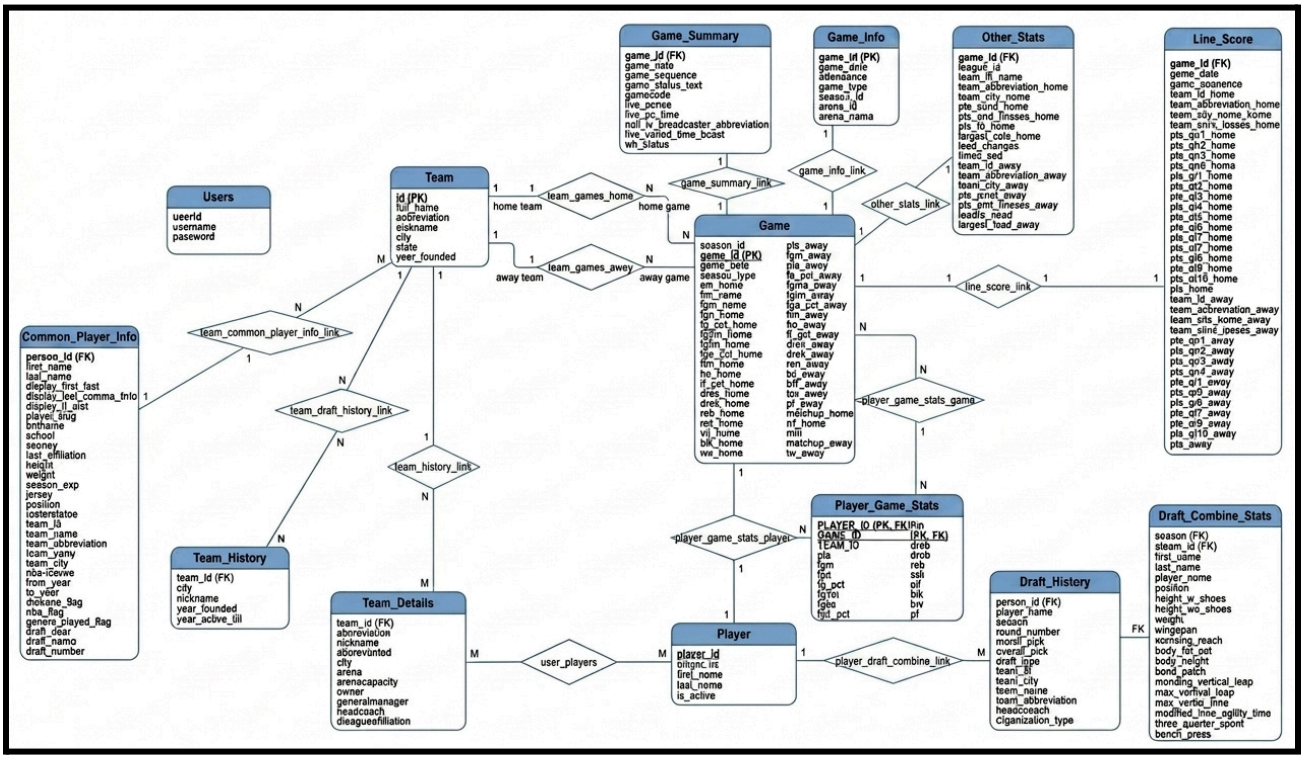
The application uses **JDBC with PreparedStatements** for all database interactions:

- Users: user authentication and profile data
- Game: NBA game records with team statistics
- Player_Game_Stats: Individual player performance per game
- Draft_History: Historical NBA draft records
- Common_Player_Info: Player biographical information
- Draft_Combine_Stats: draft combine performance metrics

Key Features

- At least 6 independent SQL queries (retrieval, insertion, updates)
- At least 3 queries with joins and aggregation** (rankings, draft search, predictions)
- 2 database indexes for performance optimization
- 1,000+ row tables populated with real NBA data (2018-19 season)

ER Model



Technology Stack

- Backend
  - Java 21
  - Spring Boot 4.0.5
    - Spring Boot Web (MVC framework)

        - Spring Boot Mustache (templating engine)
        - Spring Boot JDBC (database access)
        - Spring Security Crypto (password hashing with BCrypt)
- Database
    - MySQL (remote instance via Docker)
    - MySQL Connector/J 8.0.33
    - JDBC (Java Database Connectivity)
- Frontend
    - HTML/CSS/Javascript
    - Mustache Templating Language
- Build and Deployment
    - Apache Maven
    - Spring Boot Maven Plugin